Carbon OCR

The fast OCR engine in C++ and CUDA

Measured, Not Claimed

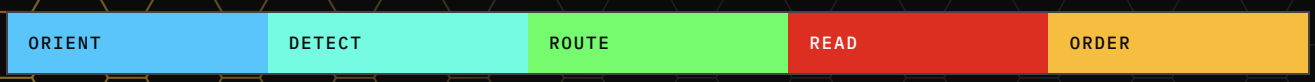
CPU and GPU · A worker per core · Many pages in flight · Every number measured

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THE MODELS

The Fullerene models, one by one

Eight small networks, 10.3 M parameters in all: what each one does, its inputs and outputs, its architecture layer by layer, its parameters and compute, and the files and metadata the engine reads. Every count comes from the layer shapes; the measured numbers are marked.



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At a glance

Five jobs, eight networks, two sets. Fullerene turns a page right, finds its lines, routes each line to its script, reads it, and puts the page in reading order. The Latin set, fullerene-3M, is 2.83 M parameters; with the other three script groups, fullerene-10M is 10.30 M.

ORIENT	`orient` · a 512 × 512 thumbnail to 12 angle bins and a vertical flag · six stride-2 convolutions · 0.34 M parameters, 255 MMAC a page
DETECT	`det` · the page at 1024 × 800 to four maps at stride 2 · an encoder to stride 32 and a 32-channel FPN · 0.44 M, 2.09 GMAC a page
ROUTE	`script` · a 32 × 512 crop to 6 groups · the crop at half size, four convolutions and a pooled linear layer · 0.06 M, 3.7 MMAC a line
READ	`rec-latin, -rtl, -indic, -cjk` · a 32 × W crop to class scores per 4 px · a RepVGG backbone, two mixing blocks around a 2× upsample, a CTC head · 1.41 to 4.37 M, 195 to 565 MMAC a line
ORDER	`order` · 16 features per line to reading order, 13 block labels and table structure · three transformer layers and five pairwise heads · 0.48 M, 22 MMAC a page

2.83 M
parameters in fullerene-3M, the Latin set: orient, det, script, rec-latin, order

10.30 M
parameters in fullerene-10M, every script group; rec-cjk's 4.37 M is the largest

10.8
GMAC to read a 40-line Latin page, recognition 77% of it

128
CTC frames in a 512-px line: one per 4 px, from a 2× upsample with no parameters

3
precisions a model ships in: FP32 for CPUs, FP16 for TensorRT, INT8 for the engine's own kernels

STATUS, STATED PLAINLY

The Fullerene weights are not released yet. The engine is built and reads a set as soon as it ships: the files of a set, the metadata the engine reads from them and the precisions each backend runs are specified here.

HOW THE NUMBERS ARE COUNTED

Parameters and MACs come from the layer shapes, with batch norm folded as in the shipped files. Measured numbers are marked with their machine.



PART I

The family

Eight networks in two sets, and the rules every model shares.

01 The family

02 What every model shares

01 The family

MODEL	JOB	IN	OUT	PARAMETERS	COMPUTE PER CALL	SET
`orient`	page orientation	a 512 × 512 thumbnail	12 angle bins and a vertical class	0.34 M	255 MMAC a page	both
`det`	line detection	the page at 1024 × 800	four maps at stride 2	0.44 M	2.09 GMAC a page	both
`script`	script group of a line	a 32 × 512 crop	one of 6 groups	0.06 M	3.7 MMAC a line	both
`rec-latin`	read Latin, Cyrillic and Greek lines	a 32 × W crop	800 class scores per 4 px	1.51 M	208 MMAC a 512-px line	both
`rec-rtl`	read Arabic-script and Hebrew lines	a 32 × W crop	400 class scores per 4 px	1.41 M	195 MMAC a line	10M
`rec-indic`	read Indic and Southeast Asian lines	a 32 × W crop	1,500 class scores per 4 px	1.69 M	231 MMAC a line	10M
`rec-cjk`	read Chinese, Japanese and Korean lines	a 32 × W crop	12,000 class scores per 4 px	4.37 M	565 MMAC a line	10M
`order`	reading order, block labels, table structure	16 features per line	an order, 13 block labels, table cells	0.48 M	22 MMAC a 40-line page	both

Two sets. A set is shipped and versioned whole, and named by its parameter count rounded to the nearest million, halves up:

- **`fullerene-3M`**, the Latin set, 2.83 M parameters: `orient`, `det`, `script`, `rec-latin` and `order`.
- **`fullerene-10M`**, every script, 10.30 M parameters: the Latin set plus `rec-rtl`, `rec-indic` and `rec-cjk`.

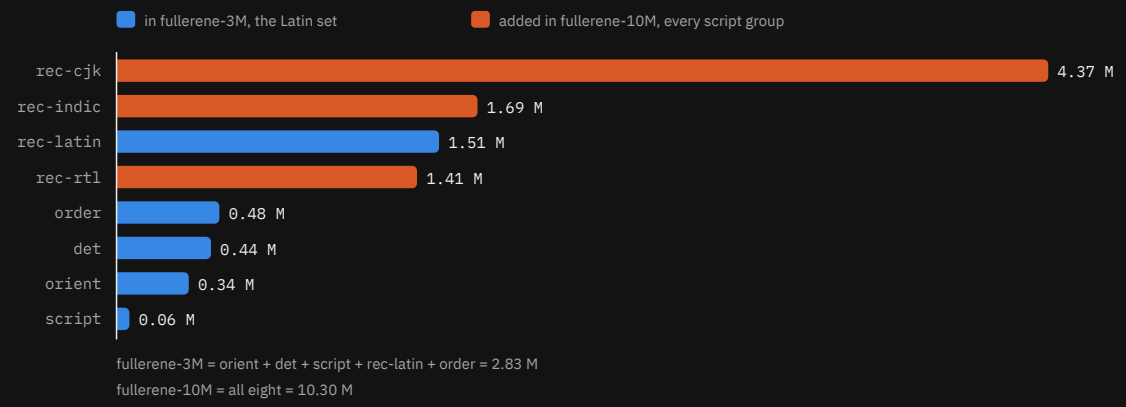


Figure 1. Parameters per Fullerene model. The Latin set, fullerene-3M, is 2.83 M; every script group, fullerene-10M, is 10.30 M.

What a page costs. A typical page is US Letter at 200 dpi (1700 × 2200 px) with 40 lines, each crop 512 px wide after its padding:

STAGE	LATIN PAGE	CJK PAGE
`orient`, once	0.25 GMAC	0.25 GMAC
`det`, once	2.09 GMAC	2.09 GMAC
`script`, 40 lines	0.15 GMAC	0.15 GMAC
the recognizer, 40 lines	8.33 GMAC (`rec-latin`)	22.59 GMAC (`rec-cjk`)
`order`, once	0.02 GMAC	0.02 GMAC
total	**10.84 GMAC**	**25.10 GMAC**

Recognition is 77% of a Latin page and 90% of a CJK page, which is why the recognizers get the most design effort and the first GPU kernels (the fused CTC head). Decoding (connected components, CTC's collapse, the order chain), rendering and image decoding are not counted: they are C++ and CUDA, and timed per stage in each page's result.

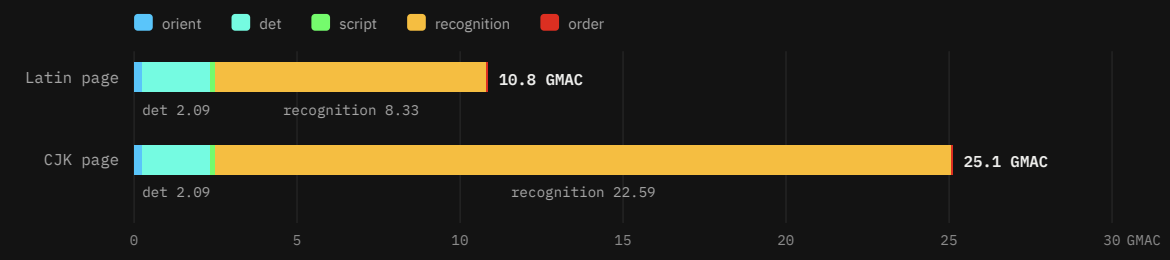


Figure 2. Compute per page by stage, in GMAC, for a 40-line page. Recognition is 77% of a Latin page and 90% of a CJK page.

The engine does not run all of this yet: it reads every line of a page with one recognizer (`rec-latin` by default), so `script` and the routing it drives are still to come (ENGINE.md, section 1).

02 What every model shares

- **Ink maps in.** The image models read float32 ink maps, NCHW with one channel: 1 − gray / 255, so ink is 1 and paper 0. The engine makes them from 8-bit gray pages on the CPU, or on the GPU (`cuda::ink_map`) in FP16 for TensorRT's files.
- **Plain operations.** The convolutional networks use ReLU, never HardSwish, and their channel counts are multiples of 16, for tensor cores and SIMD lanes.
- **Folded.** Each convolutional stage is a RepVGG-style block: a 3×3, a 1×1 and an identity branch, each with batch norm, summed. In the shipped files every block is folded into one 3×3 convolution with a bias, so the graph is a chain of plain convolutions, which TensorRT, ONNX Runtime and the engine's native backend all fuse well. The parameter counts here are the folded ones.
- **CPU and GPU pull in different directions.** Depthwise convolutions are cheap on a CPU and memory-bound on a GPU; dense 3×3 convolutions are the reverse. So the networks are dense 3×3 convolutions, with depthwise ones only where they run on one row (the recognizers' mixing blocks) or on 16 channels (the detector's head).

- **ONNX with free dimensions.** Every model is ONNX opset 17, its batch free, and its width (the recognizers), its height and width (`det`) or its line count (`order`) free too.

Fixed numbers.

QUANTITY	VALUE	WHY
render resolution	200 dpi	enough for 6 pt text; a Letter page is 1700 × 2200 px
`det` input	long side 1024, padded to a multiple of 32	a 10 pt line is about 14 px high at this scale
`det` output stride	2	lines 6 px apart stay separate
line crop height	32 px	one height for every recognizer
crop padding	0.2 of the line's height above and below, 0.3 at each end	ascenders and descenders survive the detector's box errors
CTC frame	4 px of crop width	a Latin letter is about 9 px wide in a 32-px crop, and CTC needs a frame per character plus a blank between repeats
width buckets	32 to 2048 px, each at most 1.3 times the last	padding stays small, and a batch meets few shapes
turn threshold	the orientation model's own, else 0.95	a wrong turn loses the page



PART II

Page models

Orientation and line detection: the two models that run once a page.

03 Page orientation: orient

04 Line detection: det

03 Page orientation: orient

Job. Before detection, find how the page is turned, so a scan fed sideways or upside down reads at all, and so a page of vertical CJK columns can go to a vertical path.

In and out. Input `x`, `[N, 1, 512, 512]`: the thumbnail's ink map, the page at the detector's scale fitted into 512 × 512 at the top left, paper around it. Output `logits`, `[N, 13]`: classes 0 to 11 are bins of 30° (class k: the text turned 30k° clockwise), and class 12 is vertical CJK columns. The engine turns a page only by a quarter turn (classes 3, 6 and 9) and only when that class's softmax probability reaches the model's turn threshold, `carbon.orient.turn_probability` in the file's metadata, else 0.95. What is left, up to 5°, is levelled by deskew, which is not a model. The vertical class is not acted on yet.

LAYER	OUTPUT (C × H × W)	PARAMETERS	MACS
3×3 convolution, stride 2, 16 channels	16 × 256 × 256	160	9.4 M
3×3 convolution, stride 2, 32 channels	32 × 128 × 128	4,640	75.5 M
3×3 convolution, stride 2, 64 channels	64 × 64 × 64	18,496	75.5 M
3×3 convolution, stride 2, 96 channels	96 × 32 × 32	55,392	56.6 M
3×3 convolution, stride 2, 128 channels	128 × 16 × 16	110,720	28.3 M
3×3 convolution, stride 2, 128 channels	128 × 8 × 8	147,584	9.4 M
global average pool	128	0	0
linear to 13 classes	13	1,677	0.0 M
total		**0.34 M**	**255 M**

Why this shape. Orientation is a property of the whole page's texture (the direction of its lines, the shape of its ascenders), not of fine detail. Six stride-2 convolutions down to an 8 × 8 grid see enough, at about a tenth of the detector's cost.

04 Line detection: det

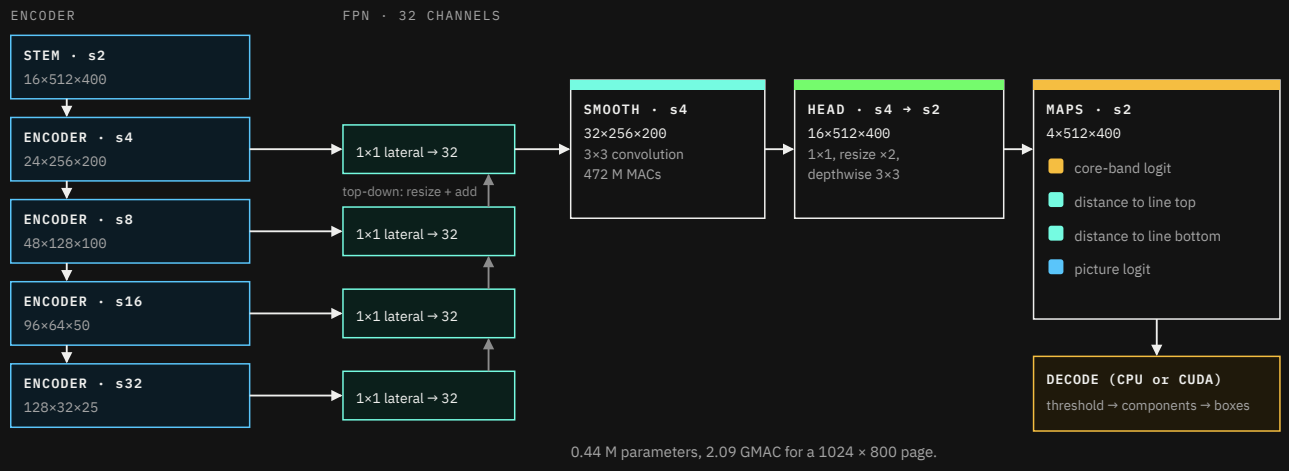


Figure 3. det, block by block: an encoder from stride 2 to stride 32, a 32-channel FPN, and four maps at stride 2. 0.44 M parameters, 2.09 GMAC for a 1024 x 800 page.

Job. Find every text line on the page, and its pictures, as boxes.

In and out. Input `x`, `[N, 1, H, W]`: the upright page's ink map at a long side of 1024, H and W padded to multiples of 32 (1024 × 800 for a Letter page). Output `maps`, `[N, 4, H/2, W/2]`, at stride 2:

MAP	VALUE
0, core	a logit: the line's core band, the middle 40% of its height, shortened at both ends by 0.15 of its height, so the bands of neighboring lines never touch
1, top	at a core pixel, the distance in input pixels from the pixel's center up to the line's top
2, bottom	the same, down to the line's bottom
3, picture	a logit: inside a picture

Map pixel (r, c) covers input rows 2r to 2r + 1 and columns 2c to 2c + 1, and is in a band when its center is. The engine's line decoder (`carbon/detect.h`, C++, no model) thresholds the core map, takes its connected components and reads each line's box from the distance maps (ENGINE.md, section 4).

BLOCK	OUTPUT (C × H × W)	PARAMETERS	MACS
stem: 3×3, stride 2	16 × 512 × 400	160	29.5 M
stride 4: 3×3 stride 2, then 3×3	24 × 256 × 200	8,688	442.3 M
stride 8: 3×3 stride 2, then 3×3	48 × 128 × 100	31,200	398.1 M
stride 16: 3×3 stride 2, then 3×3	96 × 64 × 50	124,608	398.1 M
stride 32: 3×3 stride 2, then 3×3	128 × 32 × 25	258,304	206.5 M

BLOCK	OUTPUT (C × H × W)	PARAMETERS	MACS
FPN: 1×1 laterals to 32 channels at each stride, summed top-down	32 × 256 × 200	9,600	72.1 M
FPN smoothing: 3×3 at stride 4	32 × 256 × 200	9,248	471.9 M
head: 1×1 to 16, resize ×2, depthwise 3×3, 1×1 to 4 maps	4 × 512 × 400	756	68.8 M
total		**0.44 M**	**2.09 G**

Why this shape. An encoder down to stride 32 gives the context to tell a line from a table rule or a picture's edge. A light 32-channel FPN brings that context back to stride 4, where most of the cost is (the smoothing convolution is a fifth of the model). The head goes on to stride 2, because at stride 4 lines 6 px apart would merge. The detector sees 127 px around each pixel, which is why a page can be detected in strips with 128-row halos and still give the whole page's maps, byte for byte (`carbon/strips.h`).



PART III

Line models

The script router and the four recognizers: the models that run once a line.

05 Script routing: script

06 Line recognition: the rec models

05 Script routing: script

Job. Route each line to the recognizer of its script group, so a page that mixes scripts reads each line with the right model.

In and out. Input `[N, 1, 32, W]` : the same crop the recognizers read, which the model's first operation, a 2×2 average pool with no parameters, takes to 16 px high. Output `[N, 6]` : group scores, four for the recognizers' groups (Latin, RTL, Indic and Southeast Asian, CJK) and two reserved. In the design, a line under 0.95 confidence is read by its two most likely recognizers, and the more confident reading wins. The engine does not run `script` yet; in `fullerene-3M` every line goes to `rec-latin` regardless.

LAYER	OUTPUT (C × H × W)	PARAMETERS	MACS
average pool 2×2	1 × 16 × 256	0	0
3×3 convolution, stride 2, 16 channels	16 × 8 × 128	160	0.1 M
3×3 convolution, stride 2, 32 channels	32 × 4 × 64	4,640	1.2 M
3×3 convolution, stride 2, 64 channels	64 × 2 × 32	18,496	1.2 M
3×3 convolution, stride 2 × 1, 64 channels	64 × 1 × 32	36,928	1.2 M
global average pool	64	0	0
linear to 6 groups	6	390	0.0 M
total		**0.06 M**	**3.7 M**

Why this shape. Script is visible in a few letters, so the model is about a twentieth of a recognizer and adds 2% to a line's cost. It reads the crop at half size, 16 px high, where the groups' letter shapes still differ, and pooling over the whole width makes it indifferent to the line's length.

06 Line recognition: the rec models

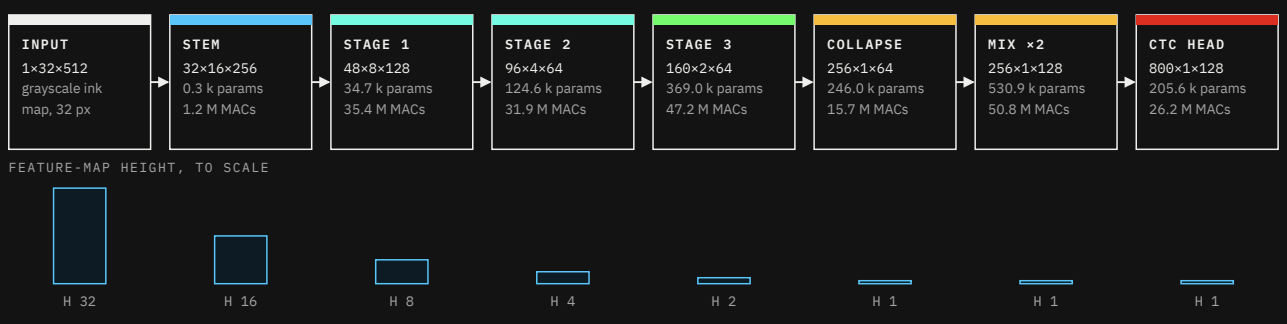


Figure 4. rec-latin, block by block, for a 32 x 512 line: the tensor shape after each block, its parameters and its MACs. 1.51 M parameters, 208 MMAC, 128 CTC frames.

Job. Read a line: turn its 32-px crop into a sequence of characters. The four recognizers share one design and differ in their vocabularies and, for CJK, their width.

In and out. Input `x`, `[N, 1, 32, W]`: a batch of crops' ink maps, W a multiple of 8 (a width bucket). Output `logits`, `[N, classes, W/4]`: a score for every class every 4 px of width, class 0 the CTC blank. The class strings are in `<rec>.vocab.json`. Greedy CTC decoding takes the best class per frame, drops blanks and collapses repeats; on a GPU it runs fused with the last layer, from the trunk's output `features`, `[N, k, W/4]` in FP16 (ENGINE.md, section 5).

`rec-latin`, layer by layer, for a 32 × 512 crop:

BLOCK	OUTPUT (C × H × W)	PARAMETERS	MACS
stem: 3×3, stride 2	32 × 16 × 256	320	1.2 M
stage 1: 3×3 stride 2, then 3×3	48 × 8 × 128	34,656	35.4 M
stage 2: 3×3 stride 2, then 3×3	96 × 4 × 64	124,608	31.9 M
stage 3: 3×3 stride (2, 1), then 3×3	160 × 2 × 64	368,960	47.2 M
collapse the height: 2×3	256 × 1 × 64	246,016	15.7 M
mixing block 1: depthwise 1×9, 1×1 to 512, 1×1 back to 256, plus the input	256 × 1 × 64	265,472	16.9 M
upsample the width 2×, nearest	256 × 1 × 128	0	0
mixing block 2: the same	256 × 1 × 128	265,472	33.8 M
CTC head: 1×1 to 800 classes	800 × 1 × 128	205,600	26.2 M
total		**1.51 M**	**208 M**

Why this shape.

- **The backbone** is dense 3×3 convolutions, which a GPU runs on its tensor cores at full rate. It halves the height and width together down to W/8 at stage 2; stage 3 halves only the height, since a line is wide and short and its characters are along the width. So stages 2 and 3 both run at W/8's positions, where the same weights cost half the MACs they would at W/4.
- **The collapse** takes the last two rows to one with a 2×3 convolution, so the model reads the full height of every letter once, at W/8.
- **The frames.** The backbone ends at a frame per 8 px, and a Latin letter is about 9 px wide in a 32-px crop: too few frames for CTC, which needs a frame per character and a blank between repeats. The first mixing block runs at W/8; then a nearest 2× upsample, with no parameters, doubles the frames to W/4, and the second block gives each frame of a pair its own context. That costs 30 MMAC a line (14% of the model), against 320 MMAC a line for a backbone kept at W/4 through stages 2 and 3. With both mixing blocks before the upsample, each pair of frames would get the same scores, which is 8-px frames again.
- **The mixing blocks** see ±48 px of the line on top of the backbone's own reach (±32 px for the first, at W/8, and ±16 px for the second, at W/4), enough to reorder Indic vowel signs locally. They are the model's only depthwise layers, used where depthwise is cheap: on one row.
- **The head** is a 1×1 convolution to the classes: 14% of `rec-latin`, and most of `rec-cjk`, which is why that one is factorized.

The four recognizers.

RECOGNIZER	CLASSES	CHANNELS (STEM, STAGE 1, 2, 3, MIXING)	HEAD	PARAMETERS	MACS AT 512 PX
`rec-latin`	800	32, 48, 96, 160, 256	1×1 to 800	1.51 M	208 M
`rec-rtl`	400	32, 48, 96, 160, 256	1×1 to 400	1.41 M	195 M
`rec-indic`	1,500	32, 48, 96, 160, 256	1×1 to 1,500	1.69 M	231 M
`rec-cjk`	12,000	32, 48, 128, 192, 320	1×1 to 192, then to 12,000	4.37 M	565 M

Compute grows linearly with the width: a 256-px line costs half as much (104 MMAC for `rec-latin`).

`rec-cjk`. Han characters are a full em wide and there are thousands of them, so `rec-cjk` is wider in its middle stages (128, 192 and 320 channels), and its head is factorized: a 1×1 down to 192 channels, then a 1×1 to the 12,000 classes. Even so, that last layer is 2.32 M parameters and 295 MMAC a line, over half the model's parameters and 52% of its compute. That makes it the first candidate for INT8, and the model that gains most from the fused CTC head on a GPU: the logits of 256 lines of 512 px would be about 750 MB in FP16, and the fused kernel never writes them.

Vocabularies.

GROUP	CLASSES	CONTENTS
Latin	800	Latin with diacritics as precomposed letters, Cyrillic, Greek, digits, punctuation, symbols, Vietnamese and Yoruba tone marks
RTL	400	Arabic, Persian, Urdu, Pashto and Kurdish letters; Hebrew with niqqud; plus Latin
Indic and Southeast Asian	1,500	Devanagari, Bengali, Tamil, Telugu, Kannada, Malayalam, Gujarati, Gurmukhi, Odia, Sinhala, Thai, Lao, Khmer, Myanmar, Georgian, Armenian, Ethiopic; plus Latin
CJK	12,000	Simplified and Traditional Chinese, kana, Hangul, plus Latin

The classes are Unicode NFC text, with canonical quotes, dashes and spaces. A right-to-left recognizer reads in visual order; putting its text in logical order (the Unicode bidirectional algorithm) is not in the engine yet. The Indic recognizer's classes are in logical order.



PART IV

Layout and beyond

Reading order and layout, and the models still to come.

07 Reading order and layout: order

08 Language models and formula models

07 Reading order and layout: order

Job. Put the page's lines in reading order (columns, sidebars, captions, footnotes), group them into labeled blocks, and recover table structure.

In. Input `x`, `[1, n, 16]`: 16 features for each of the page's n lines (`carbon::order_features`, `carbon/order.h`), from its box in page pixels and its text:

FEATURE	VALUE
0 to 3	the box, over the page's width and height
4, 5	its center, likewise
6	its width over the page's
7	the log of its height over the page's median line height
8	the log of its aspect, width over height
9	the page's median line height over the page's height
10	log(1 + its characters)
11	the share of its characters that are digits
12	1 if it starts with a list marker: a bullet or dash, or 1 to 3 digits, a letter or a roman numeral, then "." or ")" (or in brackets)
13	1 if it ends with a hyphen
14	1 if it ends a sentence: `.` `!` `?` `:` `;` or an ellipsis (CJK's too), before any closing quotes or brackets
15	1 if it starts with a capital: Latin, Greek or Cyrillic

Out. Six outputs, logits, with token 0 the page's start and tokens 1 to n its lines: `successor`, `[1, n+1, n+1]` (row i, the line after line i; column 0, the page's end); `same_block`, `table_row`, `table_col` and `table_cell`, `[1, n, n]` each, symmetric; and `labels`, `[1, n, 13]`, the label of each line's block: Text, Section-Header, Page-Header, Page-Footer, List-Group, Table, Caption, Footnote, Table-Of-Contents, Form, Picture, Formula, Other.

BLOCK	OUTPUT	PARAMETERS	MACS (40 LINES)
line features: linear 16 → 128	128 per line	2,176	0.1 M
transformer layer 1: attention (4 heads), feed-forward 256	128 per line	131,968	5.8 M
transformer layer 2: the same	128 per line	131,968	5.8 M
transformer layer 3: the same	128 per line	131,968	5.8 M
pairwise heads: successor, same block, table row, table column, table cell	41 × 41 and 40 × 40 each	81,920	4.4 M
block labels: linear to 13	13 per line	1,677	0.1 M
total		**0.48 M**	**22 M**

The start token is the embedding of a feature vector of −1s, which no line has. Each layer is a pre-norm attention and a GELU feed-forward, each with its residual, and the layer norms have no weights of their own; a last layer norm follows the third layer. Each pairwise head is a bilinear form, a 128 × 128 matrix with no bias: the successor's over every token, the others over the lines, made symmetric.

Why this shape. Reading order is a relation between lines, not a property of pixels, so the model reads the lines' geometry and a few text cues rather than the image: attention over every line of the page, then a bilinear head per relation. It costs less than recognizing one line. The engine decodes its outputs in C++ (`carbon::decode_order`): a greedy chain with repair sweeps, blocks as runs of that order, tables' cells, rows and columns from the pairwise heads, and the XY-cut's order where the chain is unsure (ENGINE.md, section 6).

08 Language models and formula models

Language models, `lm-<group>`. The design pairs each script group with a character 7-gram language model, at most 15 MB, in a packed, memory-mapped format (`lm-latin.bin` and so on). It reads again only the lines a recognizer is unsure of, those whose mean CTC peak probability is under 0.9: the recognizer hands over its top classes per frame, and a beam of 8 in C++ rescores them on CPU threads, so a smudged "rn" that could be "m" comes out as the word the language makes likelier. The language models are not counted in the sets' parameters, and the engine does not run them yet.

Formula models, `mathcls` and `math`. Planned, opt-in and off by default, in neither set: `mathcls`, a line classifier that flags the lines holding math, so the other lines never pay for it; and `math`, a small encoder-decoder that writes inline LaTeX for the flagged lines, with cached keys and values and an early stop on low confidence.



PART V

Shipping a set

The files of a set, the metadata the engine reads, the precisions and the backends that run them, and the terms used.

09 The files of a set

10 The metadata and checks the engine reads

11 Precisions and backends

12 Terms

09 The files of a set

A set is one directory. Its files are named by role and precision:

```
models/fullerene-3M/
  orient.onnx                   page orientation, FP32
  orient.fp16.onnx              FP16, for TensorRT
  orient.int8.onnx              INT8 (Q/DQ), for the native CPU backend
  orient.int8-sym.onnx          symmetric INT8, for TensorRT
  det.onnx  det.fp16.onnx  det.int8.onnx  det.int8-sym.onnx
  script.onnx  script.fp16.onnx  script.int8.onnx
  rec-latin.onnx                the recognizer, FP32
  rec-latin.vocab.json          its classes, class 0 the CTC blank
  rec-latin.fp16.onnx           FP16, for TensorRT
  rec-latin.features.fp16.onnx  its trunk, up to the last layer: TensorRT runs it,
                                then the fused CTC head
  rec-latin.head.bin            the last layer's weights, for that kernel
  rec-latin.int8.onnx           INT8 (Q/DQ), for the native CPU backend
  rec-latin.int8-sym.onnx       symmetric INT8, for TensorRT
  order.onnx                    reading order and layout, FP32
  order.native.bin              its weights, for the native CPU backend
  order.fp16.onnx  order.int8.onnx  order.int8-sym.onnx
  meta.json                     the models' contracts
  lm-latin.bin                  the language model (planned)
```

`fullerene-10M` adds the same files for `rec-rtl`, `rec-indic` and `rec-cjk`.

- **`<rec>.vocab.json`:** `{"model": "rec-latin", "blank": 0, "chars": [...]}`. The engine reads `chars`: class i is `chars[i]`, one string each, and class 0 is the blank.
- **`<rec>.head.bin`:** the recognizer's last 1×1 convolution, little-endian: the magic `CTCHEAD1`, int32 classes, int32 k (the trunk's channels: 256 for `rec-latin`, 192 for `rec-cjk`), the weight `[classes][k]` in float16, the bias `[classes]` in float32.
- **`order.native.bin`:** `order`'s weights for the engine's own kernels, little-endian: the magic `ORDERNT1`; int32 d, layers, heads, feed-forward width, features (16) and labels (13); then float32 weights, `[out][in]`: the embedding's weight and bias, each layer's attention input and output and its two feed-forward layers (weights and biases), the five pairwise forms (weights only), and the labels' weight and bias.
- **`meta.json`:** each model's contract, for tools and people: its input and output names, dtypes and shapes, its parameter count, and for a recognizer its crop rule and frame. The engine does not read it; what the engine needs is in the ONNX files and the vocabulary.

Which files each path opens:

PATH	FILES
the CPU engine (`carbon::Engine`, `Pipeline` on the CPU)	`orient`, `det` and the recognizer as `.int8.onnx` where the set has it, else `.onnx`; `det.onnx` too, for detection on several threads; `<rec>.vocab.json`; `order.native.bin`, else `order.onnx` (without either, the XY-cut)
the GPU pipeline (`Pipeline` on CUDA)	`orient.fp16.onnx`, `det.fp16.onnx`, `<rec>.features.fp16.onnx` and `<rec>.head.bin` (else `<rec>.onnx`), `<rec>.vocab.json`; the order model as on the CPU

10 The metadata and checks the engine reads

Two settings travel with the models, in their ONNX files' metadata (`metadata_props`), so a set carries what it was built for:

KEY	FILE	RANGE	WITHOUT IT
`carbon.crop.height`	the recognizer	a whole number, 8 to 512	32
`carbon.crop.pad_v`	the recognizer	0 to 2	0.2
`carbon.crop.pad_h`	the recognizer	0 to 2	0.3
`carbon.orient.turn_probability`	`orient`	0.5 to 1	0.95

The engine reads the crop rule from the first of `<rec>.onnx`, `<rec>.fp16.onnx`, `<rec>.int8.onnx` and `<rec>.int8-sym.onnx` that exists, and the turn threshold likewise from the `orient` files. A value out of its range, or one that is not a number, stops the engine with `carbon::Error`. `EngineOptions::turn_probability` overrides the set's threshold; `Engine::crop()` and `Engine::turn_probability()` report what is in use.

When it opens a set, the engine also checks that:

- every file it needs is there, and `<rec>.vocab.json` holds a `chars` list;
- the recognizer reads lines as high as its crop rule makes them (where its input height is fixed);
- the recognizer's classes, its output's second dimension (or, on the GPU, the head file's classes), equal its vocabulary's size.

When it orders a page, it checks that the order model gives six outputs.

`carbon-ocr info MODEL` prints a file's inputs and outputs, and `carbon-ocr compare MODEL` runs it on the native backend and on ONNX Runtime and prints how far apart the outputs are.

11 Precisions and backends

Each model ships in several precisions, and each backend runs the one that suits it. TensorRT 11 builds strongly typed networks, so it runs a model in its file's own precision: an FP16 file in FP16, a Q/DQ file in INT8.

FILE	PRECISION	FORMAT	RUN BY
`<m>.onnx`	FP32	plain ONNX	ONNX Runtime on the CPU; the native backend where a set has no int8 file; TensorRT for a recognizer without its trunk and head files
`<m>.int8.onnx`	INT8	Q/DQ: int8 weights per output channel, uint8 activations with zero points	the native CPU backend, the engine's default on the CPU for the convolutional networks

FILE	PRECISION	FORMAT	RUN BY
`<m>.fp16.onnx`	FP16	float16 inputs, weights and outputs	TensorRT: `orient` and `det` in the GPU pipeline
`<rec>.features.fp16.onnx`, `<rec>.head.bin`	FP16	the trunk in float16; the head's weight in float16, its bias in float32	TensorRT and the fused CTC head, the GPU pipeline's recognizer
`<m>.int8-sym.onnx`	INT8	Q/DQ: symmetric int8, biases in float	TensorRT, for `carbon-ocr bench --backend trt`; the GPU pipeline does not open it yet
`order.native.bin`	FP32	`ORDERNT1`	the native CPU backend: `order` on the engine's FP32 GEMM

`order.onnx` runs on ONNX Runtime where a set has no `order.native.bin` (or where every model is asked to run on ONNX Runtime). `order.int8.onnx`, its matrix products in Q/DQ, and `order.fp16.onnx` are for ONNX Runtime and TensorRT; the engine itself runs `order` on the CPU, from `order.native.bin` or `order.onnx`. A recognizer's whole-model `<rec>.fp16.onnx` is there for tools and other runtimes, since the GPU pipeline runs its trunk and the fused head instead.

The INT8 files are the CPU's path: the native backend's int8 kernels run a recognizer 2.5 times as fast as ONNX Runtime runs the FP32 file, while its own FP32 path is slower than ONNX Runtime's. Measured with random weights (speed does not depend on them) on an Intel Core i7-13620H and an RTX 4060 Laptop GPU, October 2026:

WHAT	MEASURED
`rec-latin`, one CPU core	8 lines of 512 px in 11.4 ms in INT8 on the native backend; 16 lines in 65 ms on ONNX Runtime 1.31 in FP32
`rec-latin`, the GPU	256 lines of 512 px in 5.1 ms in FP16 (12.8 TMAC/s), 13.6 ms in FP32 and 2.98 ms in INT8 (TensorRT 11.4)
`fullerene-3M`, one CPU core	a Latin page in 98 ms, every model included, the set's INT8 files on the native backend
`fullerene-10M`, one CPU core	a CJK page in 271 ms

12 Terms

TERM	MEANING
MAC	a multiply-accumulate: one multiplication and one addition. A layer's MACs are counted from its weights and its output's size
GMAC, MMAC	a billion, a million MACs
folded	a RepVGG block (3×3, 1×1 and identity branches with batch norm) merged into one 3×3 convolution with a bias, as in the shipped files
stride	the step of a convolution; stride 2 halves the height and width
depthwise	a convolution that filters each channel on its own: cheap in MACs, memory-bound on a GPU

TERM	MEANING
FPN	a feature pyramid network: a top-down path that adds coarse, wide context to finer maps
ink map	a model's image input: 1 – gray / 255, ink 1 and paper 0
CTC	connectionist temporal classification: reads a sequence of frames into a shorter sequence of characters, with a blank class between them
frame	one step of a recognizer's output, 4 px of crop width
logits	a classifier's raw class scores, before the softmax
the fused CTC head	the engine's CUDA kernel that runs a recognizer's last 1×1 convolution and greedy CTC decoding together, without writing the logits
Q/DQ	quantize and dequantize nodes in an ONNX graph, which carry an INT8 model's scales and zero points
a set	the Fullerene models shipped and versioned together, in one directory, named by their total parameter count

Carbon OCR

Every line read, every page in order.

Every number measured, in public.

EVERY LAYER COUNTED

Each model's parameters and MACs come from its layer shapes, and every model built matches its budget exactly.

EVERY CHOICE ARGUED

Each shape says why: the frame, the strides, the collapse, the factorized head, against what the other way would cost.

EVERY FILE SPECIFIED

Each set's files, precisions and metadata are fixed, so the engine reads a set the day it ships.

Where it stands

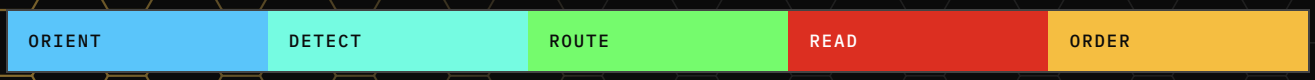
Eight small networks, 10.3 M parameters, in two sets.

Their weights come next, under OpenRAIL-AMS.

Karan Vora, Founder · Carbon Research

Carbon-OCR is the engine and **Fullerene** its models. This document lives in the repository as `docs/MODELS.md`; it is built from it, beside the engine guide.

Parameters and MACs are the design's budget, counted from the layer shapes. Measured numbers come from random weights on a laptop, 27 to 30 October 2026.



Karan Vora, Founder · Carbon Research